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REVIEW

Prevention of Post-Mastectomy Pain Syndrome: A Review of Recent Literature on Perioperative Interventions

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Abstract

Purpose of Review Up to 60% of breast cancer patients continue to experience pain three months or more after surgery, with 15 to 25% reporting moderate to severe pain. Post-mastectomy pain syndrome (PMPS) places a high burden on patients. We reviewed recent studies on perioperative interventions to prevent PMPS incidence and severity.

Recent Findings Recent studies on pharmacologic and regional anesthetic interventions were reviewed. Only nine of the twenty-three studies included reported a significant improvement in PMPS incidence and/or severity, sometimes with mixed results for similar interventions.

Summary Evidence for prevention of PMPS is mixed. Further investigation of impact of variations in dosing is warranted. In addition, promising newer interventions for prevention of PMPS such as cryoneurolysis of intercostal nerves and stellate ganglion block need confirmatory studies.

Keywords Breast cancer · Chronic Postsurgical pain · Post-mastectomy pain Syndrome

Introduction

Breast cancer is the most commonly diagnosed cancer in the United States, accounting for 15% of all cancers diagnosed in 2023 [1]. Diagnosis rates have increased steadily by 0.5% annually since 2010, and an estimated 4.1 million women with a history of breast cancer live in the United States today [2]. Early detection of breast cancer in the localized stage and regional stage has resulted in 99.3% and 86.3% survival rates, respectively, compared to 31% for breast cancer detected with metastasis [3]. Surgical removal of breast cancer has played a significant role in higher survival rates [4], potentially improving survival by 30–50% [5].

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Breast conserving surgery (BCS) rates have increased in the last decade, with one study reporting 64.6% of breast cancer patients receiving BCS in 2008 and 70.7% in 2020 [6]. Rates of bilateral mastectomies have remained relatively steady from 10.4 to 11.3% during the same period, while unilateral mastectomies decreased from 25.0–18.1% [6, 7]. Rates of prophylactic contralateral mastectomies have also increased substantially [8], further increasing the burden of mastectomies per year.

After surgery, approximately 20% of patients develop chronic postsurgical pain, defined as pain at the surgical site lasting three months or longer [9]. Surgery for breast cancer has a particularly high association with chronic postsurgical pain. This chronic pain after breast cancer surgery is termed post-mastectomy pain syndrome (PMPS). Up to 60% of patients continue to experience any pain three to six months after surgery, with 15 to 25% experiencing moderate to severe pain [10–12]. Chronic pain after breast surgery may be caused by damage to peripheral nerves, such as the intercostobrachial nerve, and tissues during surgery and may have contribution from radiation therapy and chemotherapy. Pain is in the chest wall, axilla, and/or arm, and phantom pain may be present [13]. Risk factors for PMPS include presence of preoperative pain, anxiety, depression, catastrophizing,

severe acute postoperative pain, and axillary lymph node dissection [11, 12].

Given the burden of PMPS and chronic postsurgical pain in general, interventions to prevent the transition from acute to chronic postsurgical pain have been studied, including perioperative pharmacologic and regional anesthetic techniques. In 2021, a systematic review and meta-analysis of placebo-controlled, randomized controlled trials (RCTs) evaluated perioperative pharmacologic therapies for prevention of chronic postsurgical pain [14]. Twenty-seven studies evaluated ketamine or (s)-ketamine. There were no significant differences in incidence of chronic postsurgical pain at 3 months, 6 months, and 1 year after surgery, regardless of surgical procedure. Twenty-six studies evaluated pregabalin. Though pregabalin decreased incidence of chronic pain at 3 months after cardiac surgery and total knee arthroplasty, no effects were noted for other types of surgeries, including breast cancer surgery. Eighteen studies of gabapentin showed no significant differences in incidence of chronic postsurgical pain at 3 and 6 months after surgery. Ten studies of IV lidocaine were included. In a subgroup analysis, 4 studies conducted in breast cancer surgery did not show a reduction in PMPS at 3 months; however, 2 studies conducted in breast cancer surgery showed a reduction in PMPS at 6 months compared to control (RR 0.43, 95% CI 0.23–0.80).

Regional blocks can provide good analgesia for breast surgery, reducing acute postoperative pain scores and opioid consumptions [15]. However, the evidence for prevention of PMPS is mixed. A systematic review and meta-analysis from 2019 evaluated pectoral nerve block (PECS) II for analgesia after breast cancer surgery [16]. One study in this review included long term outcome data on PMPS, with no significant differences in incidence of PMPS at 3 and 6 months after surgery between PECS II and control groups. A systematic review and meta-analysis published in 2021 included studies of single-shot or continuous paravertebral block (PVB) to decrease incidence of PMPS [17]. Twelve studies were included with a total of 2,403 patients. Six studies reported on PMPS at 3 months after surgery; 7 and 4 studies reported on PMPS at 6 and 12 months, respectively. There were no significant differences in incidence of PMPS at any of the time points between the PVB and control groups. A systematic review and meta-analysis published in 2023 evaluated analgesic effects of serratus anterior plane block (SAPB) in thoracic and breast surgery [18]. Three studies, all in breast surgery, reported on PMPS. Incidence of PMPS was significantly lower in SAPB groups compared to control groups (RR = 0.44, 95% CI 0.29–0.68, $p < 0.01$).

This review focuses on recently published literature, providing an update on the aforementioned systematic reviews as well as evaluating studies on novel perioperative interventions for prevention of PMPS.

Methods

The primary literature search was conducted in PubMed and Ovid MEDLINE. The following search terms were used: ((breast cancer) OR mastectomy)) AND ((chronic postsurgical pain) OR (postoperative pain)). Inclusion criteria for this review includes literature published in the English language between January 1, 2018, and December 4, 2023, study population of adults undergoing breast surgery for breast cancer or to lower the risk of breast cancer, and pharmacologic and regional anesthetic interventions for prevention of PMPS at 3 months or greater after surgery. Randomized controlled trials, prospective observational studies, and retrospective studies were included. Studies already included in the above systematic reviews were excluded. Teams of two independently screened the titles, abstracts, and full manuscripts. Teams of two then independently extracted data from each study. Discrepancies were reviewed by the team until consensus was reached.

Results

Pharmacologic Studies (Table 1)

Ketamine

In a double-blind, placebo-controlled RCT of adult females scheduled for elective unilateral breast cancer surgery under total intravenous anesthesia, 184 patients were randomized to receive intraoperative ketamine vs. normal saline placebo [19]. Ketamine was given as a 0.5 mg/kg bolus after induction followed by an infusion of 0.12 mg/kg/h until the end of surgery. All participants received pregabalin for two weeks postoperatively. The primary outcome was incidence of PMPS at 3 months, considered as any pain in the breast, shoulder, arm, and/or back of the surgical side. PMPS at 3 months was present in 86.9% in the control group compared to 69.0% in the ketamine group ($p = 0.005$). At 6 months, incidence of PMPS was not significantly different between the two groups. Secondary outcomes including acute pain scores and morphine consumption at 24 h postoperatively were not significantly different.

A similar RCT was conducted with participants scheduled for modified radical mastectomy (MRM) [20]. One hundred participants were randomized to receive intraoperative ketamine vs. normal saline placebo. Ketamine was given as a 0.5 mg/kg bolus followed by a 0.25 mg/kg/h infusion until the end of surgery. Chronic pain was a secondary outcome of the study; it was not stated how chronic pain was defined.

Table 1 Retrieved pharmacological intervention studies included in the current review

Study	Study Design	Sample Size	Intervention	Comparator	Chronic Pain Primary Outcome (Y/*N*)	Acute Pain Measures	Chronic Pain Measures	Acute Pain Results	Chronic Pain Results (intervention vs. comparator)	Adverse Events	Limitations
Ketamine											
Kang 2020 [19]	RCT	163 analyzed	0.5 mg/kg bolus then 0.12 mg/kg/h until end of surgery	Saline placebo	Y	NRS, morphine consumption to 24 h	Incidence of PMPS (any pain), NRS, DN4 at 3 and 6 mo	NS diff	3 mo: incidence 69% vs. 86.9% ($p=0.005$) NRS at rest 1 (0–2) vs. 1 (0–2) (NS) DN4 NS diff 6 mo: incidence: 56.1% vs. 67.9% (NS) NRS at rest 0 (0–2) vs. 0 (0–1) (NS) DN4 NS diff	NS diff, including nightmares, N/V, anxiety	All patients also received pregabalin
Zhao 2021 [20]	RCT	97 analyzed	0.5 mg/kg bolus then 0.25 mg/kg/h until end of surgery	Saline placebo	N	NRS to 48 h	Incidence of PMPS at 3 mo	NS diff in NRS at rest or movement	incidence 56.3% vs. 46.9% (NS)	NS diff, including nightmares, N/V, drowsiness	Primary outcome was QoR-40 so not powered for pain outcomes
Hasan 2023 [21]	RCT	87 analyzed	Ketamine 0.5 mg/kg + magnesium sulfate 50 mg/kg bolus then ketamine 0.12 mg/kg/h + magnesium sulfate 8 mg/kg/h for the first 24 h post-operatively	Ketamine 0.5 mg/kg bolus then ketamine 0.12 mg/kg/h for the first 24 h post-operatively + saline placebo	N	Morphine consumption, NRS to 24 h	Incidence of chronic neuropathic pain at 3 mo (DN4 ≥ 4)	NS diff in NRS at rest or movement; morphine consumption lower in intervention group	incidence neuropathic pain:16.3% vs. 20.5% (NS)	NS diff in N/V	Absence of placebo group for ketamine effects
Lidocaine											
Kendall 2018 [23]	RCT	150	1.5 mg/kg bolus then 2 mg/kg/h until 1 h after placement of the last suture. Max dose 1200 mg	Saline placebo	Y	NRS, opioid consumption to 24 h	Incidence of PMPS at 3 and 6 mo according to IMMPACT criteria (see text)	NS diff	3 mo: incidence PMPS 5% vs. 10% (NS) BPI total pain score: 1 (0–3) vs. 1 (0–3) (NS) 6 mo: incidence PMPS 3% vs. 3% (NS) BPI total pain score: 0 (0–1) vs. 0.5 (0–2) (NS)	None reported	High lost to follow up at 3 and 6 mo

Table 1 (continued)

Study	Study Design	Sample Size	Intervention	Comparator	Chronic Pain Primary Outcome (Y/N)	Acute Pain Measures	Chronic Pain Measures	Acute Pain Results	Chronic Pain Results (intervention vs. comparator)	Adverse Events	Limitations
Toner 2021 [24]	RCT	150	1.5 mg/kg bolus then 2 mg/kg/h infusion intraoperatively Subcutaneous infusion 1.33 mg/kg/h for up to 12 h in recovery ward	Saline placebo, equivalent dosage and regimen	N	None	Incidence of PMPS at 3 and 6 mo	N/A	3 mo: any pain 20% vs. 25% (NS); moderate or severe pain 9% vs. 13% (NS) 6 mo: any pain 27% vs. 30% (NS); moderate or severe pain 12% vs. 12% (NS)	Safety outcomes were similar. 1 in lidocaine group met stopping criteria (visual disturbance, dizziness, metallic taste); 2 in placebo (bradycardia, visual disturbance and metallic taste).	Pilot study; lack of formal sample size calculation
Pregabalin											
Mir 2020 [25]	RCT	117	75 mg1 h before surgery, 75 mg every 12 h for 15 days	Placebo pills		NRS to 1 mo	NRS at 6 mo	At rest: 24 h: 1.47 (1.28) vs. 2.43 (2.0) ($p<0.01$); 48 h: 0.76 (0.8) vs. 2.20 (1.98) ($p<0.001$); 1 wk: 0.71 (1.1) vs. 1.57 (1.70) ($p<0.001$); 1 mo: 1.51 (1.3) vs. 1.13 (1.0) (NS) With abduction: 24 h: 2.15 (1.70) vs. 3.01 (2.27) ($p<0.01$); 48 h: 1.31 (0.94) vs. 2.67 (2.0) ($p<0.001$);1 wk: 1.19 (1.24) vs. 1.85 (1.8) ($p=0.01$); 1 mo: 1.95 (1.50) vs. 1.69 (1.37) (NS)	6 mo: NRS at rest 1.53 (1.3) vs. 1.4 (1.3) (NS) NRS with movement 2.01 (1.60) vs. 1.84 (1.52) (NS)	N/V more common in placebo; dizziness, blurred vision, headache similar	Lack of sample size calculation; published as commentary

Table 1 (continued)

Study	Study Design	Sample Size	Intervention	Comparator	Chronic Pain Primary Outcome (Y/N)	Acute Pain Measures	Chronic Pain Measures	Acute Pain Results	Chronic Pain Results (intervention vs. comparator)	Adverse Events	Limitations
Wang 2023 [26]	RCT	90	(EP group) pregabalin 150 mg 1 h before surgery then 150 mg twice daily for 7 days Starting in PACU and for total 48 h, via PCA pump: 100 µg sufentanil + 1.25 mg/kg (s)-ketamine + 4 mg tropisetron in 100 mL saline solution	Placebo pills and saline placebo in PCA	Y	Opioid consumption to 48 h and NRS at rest and during coughing to POD7	Incidence PMPS (any pain) at 3 and 6 mo	Opioid consumption lower in EP to 48 h ($p<0.05$) NRS lower in EP at rest up to POD3 NRS with coughing lower in EP up to POD7	3 mo 14.3% vs. 46.3% ($p<0.01$) 6 mo 7.1% vs. 31.7% ($p<0.01$)	4% with hallucinations and nightmares in intervention group. NS diff in N/V, dizziness, pruritus.	No data on actual PCA usage

Results are reported as median (IQR) or mean (SD)

BPI: Brief Pain Inventory; DN4: Douleur Neuropathique 4 questionnaire; NRS: numeric rating scale (0–10); NS diff: no significant difference; N/V: nausea and vomiting; PACU: post-anesthesia care unit; PCA: patient-controlled analgesia; PMPS: post-mastectomy pain syndrome; POD: post-operative day; RCT: randomized controlled trial

At 3 months, 46.9% of patients in the ketamine group had chronic pain compared to 56.3% of controls ($p=0.36$). The primary outcome Quality of Recovery (QoR) score on post-operative day 1 as well as acute pain scores were not significantly different between groups.

Another RCT compared addition of magnesium sulfate (Mg) with ketamine vs. ketamine alone, testing the hypothesis that the medications, both antagonists with different active sites for N-methyl-D-aspartate (NMDA) receptors, in combination would improve analgesic effects [21]. Forty-three patients received a bolus of 0.5 mg/kg of ketamine and 50 mg/kg of Mg with induction of anesthesia, then 0.12 mg/kg/h ketamine and 8 mg/kg/h Mg infusions for the first 24 h post-operatively. Meanwhile, 44 participants in the comparator group received a bolus of 0.5 mg/kg of ketamine then a 0.12 mg/kg/h ketamine infusion and 8 mg/kg/h of saline infusion for the first 24 h after surgery. Secondary outcomes included chronic neuropathic pain at 3 months postoperatively, defined as score ≥ 4 on the Douleur Neuropathique 4 (DN4) questionnaire. No significant differences were reported in the incidence of chronic neuropathic pain at 3 months, with incidence of 16.3% in the ketamine and Mg group compared to 20.5% in the ketamine alone group.

A single-center cohort study from 2021 is not included due to inconsistencies in reporting; it is unclear if it was retrospective or prospective study [22].

Lidocaine

In a double-blind, placebo-controlled RCT, intraoperative systemic lidocaine was found to have no effect on chronic pain compared to a saline placebo 3 and 6 months following breast cancer surgery [23]. The lidocaine group received a 1.5 mg/kg bolus at induction then a 2 mg/kg/h infusion until one hour following placement of the last suture, with a maximum dose of 1200 mg. Chronic pain was defined by the Initiative on Methods, Measurement, and Pain Assessment in Clinical Trials (IMMPACT) criteria: presence or absence of pain; average daily pain score, pain at rest, and pain with movement ≥ 4 on Numeric Rating Score (NRS); McGill pain questionnaire score ≥ 12; and BPI interference ≥ 4. Using this definition, the incidence of chronic pain between groups was not statistically significant different; 5% of the lidocaine group reported chronic pain vs. 10% of the placebo ($p=0.35$) at 3 months. At 6 months the incidence was the same in both groups at 3%.

A pilot multicenter RCT also tested efficacy of lidocaine vs. saline placebo on chronic pain at 3 and 6 months after breast cancer surgery [24]. Participants received an intravenous bolus of 1.5 mg/kg lidocaine and then a 2 mg/kg/h infusion intraoperatively. Patients then received a subcutaneous infusion of lidocaine at a rate of 1.33 mg/kg/h for up to 12 h postoperatively. Primary outcomes were safety

events, lidocaine serum levels, and enrollment rates. Secondary outcomes were any chronic pain reported at 3 and 6 months, with moderate to severe pain defined by the worst pain in the last 24 h ≥ 4. There was no statistically significant difference between incidence of chronic pain in the lidocaine intervention and placebo control. Pain of any intensity occurred in 20% and 25% at 3 months and at 27% and 30% at 6 months for the lidocaine and placebo groups, respectively. Moderate to severe intensity pain occurred in 9% and 13% at 3 months and at 12% and 12% at 6 months for the lidocaine and placebo groups, respectively.

Pregabalin

In 2020, a double-blind RCT evaluated use of pregabalin to prevent acute and chronic pain in patients undergoing mastectomy or lumpectomy with or without axillary node dissection [25]. Seventy-five participants were randomized to the pregabalin arm and received 75 mg pregabalin before surgery, followed by 75 mg every 12 h for 15 days following the surgery. Seventy-five participants received placebo pills and followed the same protocol. Although the pregabalin group reported significantly lower pain scores up to 1 week after surgery, at 6 months after surgery there were no significant differences between the pregabalin group verbal numeric rating scale (VNRS) at rest (1.53 (SD 1.36) for pregabalin vs. 1.40 (SD 1.30) for placebo ($p = 0.542$)) nor with arm abduction (2.01 (SD 1.60) for pregabalin vs. 1.84 (SD 1.52) for placebo ($p = 0.499$)).

Another double-blind RCT investigated the combination of pregabalin and (s)-ketamine vs. placebo pills and saline placebo on the incidence of chronic pain at 3 and 6 months after breast cancer surgery [26]. Patients in the intervention group received 150 mg pregabalin 1 h before surgery and twice daily for 7 days after surgery; in addition in the post anesthesia care unit (PACU), patients used a patient controlled analgesia (PCA) pump with sufentanil and (s)-ketamine (1.25 mg/kg per 100 ml saline). PMPS was defined as pain at incision site with NRS score at rest ≥ 1. At 3 months, 14.3% of the intervention group experienced chronic pain vs. 46.3% of placebo patients ($p < 0.01$). At 6 months, there continued to be a significant difference between groups with 7.1% experiencing chronic pain in the intervention group compared to 31.7% in placebo ($p < 0.01$).

Regional anesthesia studies (Table 2)

PVB

One RCT studied the effect of a preoperative PVB on chronic pain after breast cancer surgery [27]. The study included 352 patients, with 178 in the intervention group and 174 in the sham group. The primary outcome was the incidence of chronic pain, defined by Brief Pain Inventory (BPI) average pain in the past 24 h ≥ 3. At 3, 6, and 12 months, there was no significant difference in the proportion of patients reporting chronic pain between the intervention and control groups.

One RCT tested the hypothesis that cancer recurrence is lower in patients receiving regional anesthesia with PVB and propofol compared to general anesthesia with volatile agents [28]. PMPS at 6 and 12 months were secondary outcomes. Pain was assessed with the modified BPI and Neuropathic Pain Questionnaire Short Form. Pain was not different between the groups, with 52% experiencing any pain and 10% reporting neuropathic pain in the PVB at 6 months. Conversely, 52% of patients in the general anesthesia group experienced any pain at 6 months, with 28% reporting neuropathic pain. At 12 months, 28% in the PVB group had any pain vs. 27% in the general anesthesia group.

Another RCT assessed the effectiveness of multilevel single-shot PVB vs. no block in reducing PMPS in 218 patients undergoing unilateral breast surgery [29]. The primary outcome was the incidence of PMPS at 6 months, with secondary outcomes including acute pain scores and the incidence of neuropathic pain at 6 and 12 months. At 6 months, 12.5% of the PVB group reported chronic pain compared to 24% in the control group ($p = 0.031$). At 12 months, the incidence was 9.9% in the PVB group vs. 19.4% in the control group ($p = 0.055$).

SAPB

One RCT studied the effect of ultrasound-guided SAPB on acute and chronic pain among 60 patients undergoing MRM [30]. SAPB was compared to a control group receiving saline injected subcutaneously. The primary outcome was PMPS at 6 months. In the intervention cohort, 6% experienced chronic pain with a Visual Analogue Scale (VAS) ≥ 4, and 83% reported some level of pain. In the control group, 16% reported a VAS ≥ 4 ($p = 0.228$), with 83% experiencing any pain.

In 2021, an RCT compared SAPB against a sham block with saline in patients undergoing unilateral MRM [31]. The primary outcome was any chronic pain at 3 months. At 3 months, 25.6% of the SAPB group had chronic pain vs. 51.7% in the sham group ($p < 0.001$); at 6 months, 18.9% of the SAPB group had chronic pain vs. 41.6% in the sham group ($p = 0.001$). Overall, pain scores were low, with the SAPB group reporting median pain scores of 0 (IQR 0-0.25) and the sham group 2 (IQR 0-3.5) at 3 months.

In 2022, another RCT randomized 65 women undergoing MRM to receive either a SAPB (32 patients) or no

Table 2 Retrieved regional anesthesia intervention studies included in the current review

Study	Study Design	Sample Size	Intervention	Comparator	Chronic Pain Primary Outcome (Y/*N*)	Acute Pain Measures	Chronic Pain Measures	Acute Pain Outcomes	Chronic Pain Outcomes (intervention vs. comparator)	Adverse Events	Limitations
Paravertebral Block (PVB)											
Sessler 2019 [28]	RCT	2131 analyzed	PVB with propofol sedation (PVB typically T1-T5 with 5 ml long-acting local anesthetic at each level)	General anesthesia with volatile agent	N	Opioid consumption to 48 h	Incidence of PMPS at 6 and 12 mo	NS diff	6 mo: incisional pain incidence 52% vs. 52% group (NS) Neuropathic pain incidence: 10% vs. 10% (NS) 12 mo: incisional pain incidence 28% vs. 27% (NS) Neuropathic pain 7% vs. 7% (NS)	N/V higher in general anesthesia group	PVB not standardized across sites
Lin 2020 [29]	RCT	214 analyzed	Multilevel single-shot PVB from T2 to T5 with 5 mL of 0.5% ropivacaine at each dermatome	No block	Y	NRS to 48 h	Incidence of PMPS at 6 and 12 mo	NRS scores at rest and with movement lower in PVB group	6 mo: 12.5% vs. 24% ($p = 0.031$) 12 mo: 9.9% vs. 19.4% (NS)	NS diff, including hypotension, bradycardia, dizziness, N/V	No sham injection
Albi-Feldzer 2021 [27]	RCT	352 in ITT sample	Single-shot PVB at T2 with 0.35 ml/kg 0.5% ropivacaine, max 30 ml	Sham PVB with saline	Y	VAS to 48 h	Incidence of PMPS at 3, 6, and 12 mo (BPI average pain in past 24 h ≥ 3)	Higher pain in control group 0 to 2 h, NS diff at other time points	3 mo: incidence 52.2% vs. 47.7% (NS) 6 mo: incidence 58.4% vs. 45.4% (NS) 12 mo: incidence 59% vs. 53.4% (NS)	9 with Horner syndrome in PVB group	Concern for blinding integrity as block level assessed prior to induction
Serratus Anterior Plane Block (SAPB)											

Table 2 (continued)

Study	Study Design	Sample Size	Intervention	Comparator	Chronic Pain Primary Outcome (Y/*N*)	Acute Pain Measures	Chronic Pain Measures	Acute Pain Outcomes	Chronic Pain Outcomes (intervention vs. comparator)	Adverse Events	Limitations
Qian 2021 [31]	RCT	179 analyzed	SAPB with 30 mL of 0.5% ropivacaine	Sham block with saline	Y	NRS, Morphine consumption to 24 h	Incidence of PMPS (any pain) at 3 and 6 mo	Lower AUC of NRS scores at rest and with movement in SABP group. Intervention group had 6 mg (4–8 mg) morphine consumption vs. 29 mg (28–31 mg) in control group ($p < 0.001$)	3 mo: incidence PMPS 25.6% vs. 51.7% ($p < 0.001$) 6 mo: incidence PMPS 18.9% vs. 41.6% ($p = 0.001$)	Lower N/V in SAPB group	Single center study
Chai 2022 [32]	RCT	65	SAPB with 30 mL of 0.375% ropivacaine	No block	N	Sufentanil use to 24 h	Incidence of PMPS as identified by the ID-Pain scale ≥ 2 at 3, 6, and 9 mo	Intervention group used 26.3 (1.9) mcg sufentanil vs. 30.2 (3.9) mcg in control group ($p < 0.001$)	3 mo: 15.63% vs. 39.39% ($p = 0.032$) 6 mo: 12.50% vs. 33.33% ($p = 0.046$) 6 mo: 12.50% vs. 24.24% (NS)	NS diff in use of vasoactive medications	Single-blind design; single-center; small sample size; no sham
Fuzier 2022 [33]	Cohort study	137	SAPB with 20 mL of 0.375% ropivacaine	N/A	Y	Percentage of patients requiring morphine in PACU (given when NRS > 3)	Incidence of PMPS at 3 mo	18%	31.4%	Not reported	Lack of control group; single-center design
Sulak 2022 [30]	RCT	60	SAPB with 20 mL of 0.25% ropivacaine	Control group receiving saline injected subcutaneously	Y	VAS and opioid consumption to 24 h	Incidence of PMPS at 6 mo	Higher VAS in control group to 4 h, NS diff at other time points. Opioid use significantly lower in SAPB group for 1st 24 h	incidence of any pain: 83% vs. 83% (NS) incidence VAS ≥ 4: 6% vs. 16% (NS)	Not reported	Lack of patient blinding

Table 2 (continued)

Study	Study Design	Sample Size	Intervention	Comparator	Chronic Pain Primary Outcome (Y/*N*)	Acute Pain Measures	Chronic Pain Measures	Acute Pain Outcomes	Chronic Pain Outcomes (intervention vs. comparator)	Adverse Events	Limitations
Flores 2023 [39]	Follow up study of RCT	44	Combination of SAPB with 20 mL of 0.375% ropivacaine and PECS I with 10 mL 0.375% ropivacaine blocks	No block	N	None	Neuropathic pain assessment using NRS and DN4 scores at 12 mo	N/A	NS diff in NRS. Neuropathic pain (DN4 ≥ 4) incidence: 18% vs. 9% (NS)	Not reported	No sham injection
Pectoral Nerve Block (PECS)											
Barrington 2020 [36]	RCT	104 analyzed	PECS II block with 0.45 mL/kg ropivacaine 0.465%, max 40 ml, and surgical infiltration with saline	PECS II block with saline and surgical infiltration with local anesthetic	N	None	BPI-SF global score at 3 mo	N/A	incidence chronic pain: 11.3% vs. 17.6% (NS) BPI-SF global score NS diff.	PECS group: transient neurological symptoms (5), bruising (3). Control group with SSI (1), arrythmia (1). 1 in each group with hematoma requiring repeat surgery	Primary outcome was QoR-15 so not powered for pain outcomes
Mendonca 2023 [34]	Follow up study of RCT	53	PECS II with 20 ml 0.5% ropivacaine between pectoralis minor and serratus anterior and 10 ml between pectoralis muscles	General anesthesia	Y	None	incidence of PMPS (any pain) at 3 mo	N/A	PECS II block: 22% vs. 65% ($p = 0.005$)	Not reported	High loss to follow-up from the original trial
Lizarraga 2023 [35]	RCT	34	Preoperative PECS II with 12 mL of 0.25% bupivacaine between pectoralis minor and serratus anterior muscles and 8 mL between the pectoralis muscles	Block post-mastectomy by surgeon	N	NRS in PACU and inpatient stay	NRS at 3 and 6 mo	NS diff	3 mo: mean NRS at rest 0.2 and with movement 0.7 in both groups (NS) 6 mo: mean NRS at rest 0.3 vs. 0.1, with movement 0.7 vs. 0.4 (NS)	NS diff, including wound infection, hematoma, flap necrosis, readmission, N/V, constipation	Pre-mastectomy blocks were done by anesthesiologist, post-mastectomy were done by surgeon

Table 2 (continued)

Study	Study Design	Sample Size	Intervention	Comparator	Chronic Pain Primary Outcome (Y/*N*)	Acute Pain Measures	Chronic Pain Measures	Acute Pain Outcomes	Chronic Pain Outcomes (intervention vs. comparator)	Adverse Events	Limitations
Al Ja'bari 2019 [37]	RCT	42 in ITT sample	PECS II with 20 mL 0.5% ropivacaine between the pectoralis minor and serratus anterior and 10 mL between the pectoralis muscles	No block	N	Morphine consumption and VAS to 48 h	Incidence of PMPS at 6 mo	Morphine at 48 h 6 (6.5) mg vs. 12.8 (12.5) mg ($p=0.04$). NS diff in NRS to 48 h	10.5% vs. 11.1% (NS)	None reported	No sham injection
Fujii 2019 [38]	RCT	80	PECS II with 20 mL of 0.5% ropivacaine between pectoralis minor and serratus anterior and 10 mL 0.5% ropivacaine between pectoralis muscles	SAPB with 30 mL 0.5% ropivacaine	Y	Morphine consumption and VAS to 24 h	Incidence of moderate-severe pain (≥4 on NRS) at 6 mo	Median morphine consumption:6 (3–9) mg vs. 4 (2–7) mg ($p=0.04$). NS diff in VAS at 24 h	10% vs. 33% ($p=0.03$)	Not reported	No control group without a nerve block
Other											
Beguinot 2020 [41]	RCT	95 analyzed	Continuous wound infiltration with 2 mg/mL ropivacaine 10 mL bolus then 10 mL/hr for 48 h through wound catheter	Saline placebo	N	Morphine consumption to 48 h	VAS at 3 and 6 mo	Morphine consumption: 4.8 (10.4) mg vs. 10.8 (16.5) mg (NS)	3 and 6 mo: NS diff in VAS	NS diff, including hematoma, seroma, SSI, skin necrosis, scar rupture, arm lymphedema	High technical failure
Ilfeld 2022 [43•]	RCT	60	Ultrasound-guided percutaneous cryoneurolysis of T2-T5 intercostal nerves + PVB catheter	Sham procedure + PVB catheter	N	BPI average	BPI out to 12 mo	POD 2 pain score: 0 (0-1.4) vs. 3 (2–50 ($p<0.001$). Average pain scores lower for intervention group from POD1 to POD21	1 year: incidence of any pain 3% vs. 17% ($p<0.001$)	None reported	Cryoneurolysis was performed with two different systems; blinding integrity
Genc 2022 [40]	RCT	90 analyzed	1) ESPB with 30 mL of 0.25% bupivacaine 2) PECS II with 20 mL of 0.25% bupivacaine	No block	N	Morphine consumption and VAS to 24 h	Incidence of PMPS and VAS at 3 mo	Morphine use lower in ESPB and PECS II groups vs. control, VAS lower in ESPB and PECS II vs. control	VAS: ESPB 0 vs. PECS II 0 vs. control 1 ($p=0.001$) incidence: ESPB 23% vs. PECS II 23% vs. control 70% ($p<0.001$)	Not reported	Single-blind

Table 2 (continued)

Study	Study Design	Sample Size	Intervention	Comparator	Chronic Pain Primary Outcome (Y/N)	Acute Pain Measures	Chronic Pain Measures	Acute Pain Outcomes	Chronic Pain Outcomes (intervention vs. comparator)	Adverse Events	Limitations
Salman 2021 [42•]	RCT	80	Ultrasound-guided stellate ganglion block	No block	Y	Opioid consumption to 24 h	GSNP score for PMPS	Morphine consumption 0 (0–5) mg vs. 5 (0–10) mg ($p<0.001$)	0 (0-2.25) vs. 2 (0–3) ($p=0.019$)	Pain at injection site, dysphagia, hoarseness	Single-blind; no sham injection

Results are reported as median (IQR) or mean (SD)

AUC: area under curve; BPI: Brief Pain Inventory; BPI-SF: Brief Pain Inventory (Short Form); DN4: Douleur Neuropathique 4 questionnaire; GSNP: Grading System for Neuropathic Pain; NRS: numeric rating scale (0–10); NS diff: no significant difference; N/V: nausea and vomiting; PACU: post-anesthesia care unit; PMPS: post-mastectomy pain syndrome; POD: post-operative day; RCT: randomized controlled trial; VAS: visual analogue scale

block (33 patients) alongside general anesthesia [32]. The primary outcome was 24-hour sufentanil use, and secondary outcomes included presence of neuropathic pain at 3, 6, and 9 months, defined as ID-Pain score ≥ 2. Notably, at the 3 month follow-up, 15.63% of patients in the SAPB group reported chronic pain, significantly lower than the 39.39% in the control group ($p=0.032$). At 6 months, chronic pain was reported by 12.5% of the intervention group compared to 33.33% in the control group ($p=0.046$). At 9 months, there was no significant difference between groups, with incidence of 12.5% vs. 24.24% in the intervention and control groups ($p=0.349$).

A prospective, observational cohort study explored the influence of SAPB on chronic pain 3 months after breast-conserving surgery [33]. The study included 137 patients. Chronic pain was reported by 31.4% of patients at 3 months.

PECS and Other Regional Anesthesia Techniques

In a follow-up study from a previous RCT, the efficacy of ultrasound-guided PECS II block for preventing PMPS was evaluated [34]. Women undergoing elective unilateral mastectomy had PECS II vs. no block. Long-term data was available for 53 participants. Any chronic pain longer than 3 months in the PECS II block was 22% compared to 65% in the control group ($p=0.005$).

An RCT of 34 participants investigated the impact of the timing of PECS II block, preoperatively vs. post-mastectomy on postoperative pain in patients undergoing bilateral mastectomy with immediate reconstruction [35]. The primary outcome was acute pain scores. Secondary outcomes included pain scores at 3 and 6 months. Mean pain scores at rest in the preoperative PECS II group were 0.2 and 0.3 for 3 and 6 months, respectively. The post-mastectomy PECS II group had mean pain scores at rest of 0.2 and 0.1 for 3 and 6 months, respectively. The differences were not significantly different.

A multicenter RCT evaluated the quality of recovery after breast surgery, comparing the efficacy of PECS II with surgical infiltration [36]. A total of 104 patients participated, with 53 receiving the PECS II block and 51 undergoing surgical infiltration. The primary outcome was the QoR-15 at 24 h, and the secondary outcome was the BPI-SF global score at 3 months. In the PECS II group 11.3% of patients had chronic pain vs. 17.6% in the surgical infiltration group. These results were not statistically significant.

An RCT in Switzerland assessed the efficacy of the PECS II block for pain management in radical mastectomy [37]. A total of 42 patients completed the study, with 20 in the PECS II group and 22 in the control group. The primary outcome was the cumulative morphine dose at 24 h. Secondary outcomes included surgical site pain that persisted for at least 2 months. The study found that 10.5% of patients in

the intervention group reported chronic pain vs. 11.1% in the control group; these results were statistically insignificant.

Another RCT randomized 80 women undergoing mastectomy for breast cancer into two groups, with 40 patients receiving PECS II and another 40 receiving SAPB [38]. The primary outcome was the incidence of moderate to severe pain, defined as a score ≥ 4 on the NRS, assessed 6 months. Only 10% of the PECS II block experienced chronic pain compared to 33% in the SAPB group ($p = 0.03$).

One RCT divided 44 patients scheduled for mastectomy into two groups: one received a combination of SAPB and PECS I blocks (22 patients), while the other received only general anesthesia (22 patients) [39]. The primary outcome focused on neuropathic pain assessment using NRS and DN4 scores at 12 months. The results showed no significant difference in NRS pain scores between the groups at one year, with 18% of the intervention group and 9% of the control group displaying probable neuropathic pain based on DN4 scores.

Another RCT allocated 90 patients undergoing unilateral partial mastectomy into three groups: 30 received a PECS II, 30 received an Erector Spinae Plane Block (ESPB), and 30 formed the control group, receiving no blocks [40]. The primary outcome was total morphine consumption at 24 h, while secondary outcomes included VAS scores at rest and during abduction at 24 h and 3 months. At 3 months, both ESPB and PECS II groups reported significantly lower VAS scores at rest compared to the control group, with PECS II and ESPB at 0 (IQR 0–1) and the control at 0 (IQR 0–2, $p = 0.008$). Chronic pain incidence at 3 months was 23% in both PECS II and ESPB groups vs. 70% in the control group ($p < 0.001$).

Additional Studies of Interventional Techniques

An RCT assessed the effectiveness of continuous wound infiltration with ropivacaine in reducing postoperative pain and morphine consumption after unilateral total mastectomy [41]. The study enrolled 95 patients, 46 receiving ropivacaine and 49 receiving a saline placebo. The primary outcome was morphine consumption at 48 h. Secondary outcomes included pain intensity via the VAS out to 6 months. There was no statistically significant difference in pain intensity between the ropivacaine and placebo groups at 3 and 6 months.

An RCT studied ultrasound-guided stellate ganglion block for prevention of PMPS by randomizing 80 patients, with 40 in the block group and 40 in the control group [42•]. The control group received no block. The primary outcome was the proportion of PMPS at 6 months, assessed by the Grading System for Neuropathic Pain (GSNP), with PMPS considered to be grade 3 or 4 GSNP. Secondary outcomes were NRS. The proportion of PMPS at 6 months was 30% in the stellate ganglion block group vs. 62.5% in the control group ($p = 0.004$). The intervention group reported pain scores with a median of 0 (IQR 0–3]) vs. 2 (IQR 0–3) in the control group at 3 months ($p = 0.024$). At 6 months the median NRS was 0 (IQR 0-2.25) vs. 2 (IQR 0–3) in the stellate ganglion and control groups, respectively ($p = 0.019$).

One RCT randomized 60 participants undergoing mastectomy to preoperative ultrasound-guided percutaneous cryoneurolysis of T2-T5 intercostal nerves vs. a sham procedure to assess the effect in managing post-mastectomy pain [43•]. All participants also had PVB catheters with postoperative infusion until the morning of discharge. Notably, on the second postoperative day, the intervention group reported significantly lower pain scores compared to the control group (primary outcome). At the 3, 6, and 12 months, the incidence of any pain was significantly lower in the cryoneurolysis group compared to the sham group ($p < 0.001$). At 1 year, the intervention group reported a 3% incidence of chronic pain against 17% in the control group.

Discussion

Many of the studies included in this review employed approaches used in the past. Similar to a recent systematic review on perioperative pharmacologic therapies [14], studies published since then continue to show mixed results of pharmacotherapies for the prevention of PMPS.

Similarly, there is mixed evidence for efficacy of regional anesthesia techniques for prevention of PMPS. A review of relevant anatomy is paramount to understand the potential benefits of regional anesthesia in preventing PMPS. The breast receives sensory innervation primarily from the anterior and lateral cutaneous branches of the intercostal nerves spanning from T1 to T6. These nerves arise from the intervertebral foramina and split into dorsal and ventral rami. The ventral rami travel along the costal groove beneath each rib forming the intercostal nerves. The intercostobrachial nerve, a branch of T2, is of particular concern during axillary lymph node dissection and can contribute to PMPS if damaged. Supraclavicular nerves may innervate a portion of the superior breast. The lateral pectoral nerve and the medial pectoral nerve, branches of the brachial plexus, innervate the chest wall muscles and contribute to myofascial pain, especially if submuscular tissue expander or implant is used for reconstruction. The long thoracic nerve and the thoracodorsal nerve, also branches of the brachial plexus, are also of importance during axially lymph node dissection. PVBs block the intercostal nerves but miss branches of the brachial plexus and the supraclavicular nerves. PECS II and SABP block branches of the brachial plexus and lateral

cutaneous branches of the intercostal nerves but may not cover the medial or superior portions of the breast [44]. Future research may explore the potential of blocking additional nerves not currently targeted by standard regional anesthesia techniques.

Newer interventional techniques are also being studied. A particularly interesting finding in our review was the effectiveness of cryoneurolysis of intercostal nerves combined with PVB in managing PMPS, as demonstrated in the study by Ilfeld et al [43•]. Pain relief from cryoneurolysis typically lasts weeks to months. In this study, the treatment showed a substantial reduction in PMPS at one year post-surgery, with the intervention group reporting a 3% incidence compared to 17% in the control group. No patients in the intervention group experienced phantom breast pain, while 19% in the control group did at one year. While the results are promising, the study had a small sample size and was a single-center design, necessitating additional research. In addition, wider adoption of this technique could be limited by need for special equipment and time for cryoneurolysis application (~40–50 min for unilateral mastectomy). A single study also showed promising effects of preoperative stellate ganglion block in reducing the incidence of PMPS at 6 months after surgery [42•]. Stellate ganglion block has typically been used for sympathetically maintained chronic pain. Small studies have shown benefit of stellate ganglion for acute postoperative pain [45, 46], and the study by Salman et al. included here also showed decreased pain scores in the acute postoperative period through 6 months. Limitations of this study included lack of sham procedure and single-center design. Use of stellate ganglion block is likely limited to unilateral surgeries due to risks of bilateral blockade, such as stridor.

This review has several limitations. It was not conducted as a systematic review. Additionally, we did not review different surgical and behavioral treatments. Another limitation in our analysis is the inconsistency across the clinical investigations regarding the definition and measurement of chronic pain which hinders comparison between studies. Various studies utilized different definitions, outcomes, and timelines for reporting PMPS. For example, at least 3 months of pain was almost universally accepted as the time threshold to be considered chronic pain; however, some studies differed, defining chronic pain as persisting for at least 2 months [37] or at least 6 months [41]. Furthermore, there were differences in type of pain considered as PMPS. A few studies only classified pain based on duration [36, 37]. Some specified PMPS to be incisional pain whereas other studies required presence of neuropathic pain [28, 29, 41]. Some studies reported on PMPS of any intensity of pain while others only reported on PMPS with moderate to severe levels of pain [38]. The most stringent definition of pain came from the IMMPACT consortium recommendations [23, 47]. Outcome instruments also varied across studies. BPI, NRS, and VAS were used to assess pain scores. Instruments for neuropathic pain included painDETECT, ID pain scale, and Neuropathic Pain Questionnaire Short Form. While most studies used patient-reported outcomes, Salman et al. used the GSNP which requires a neurologic exam [42•, 48].

Conclusion

Prevention of PMPS remains challenging. Investigation of variations in dosing in pharmacotherapies is needed. Standardized definitions and outcome tools would help facilitate comparisons between studies. Promising newer interventions for prevention of PMPS such as cryoneurolysis of intercostal nerves and stellate ganglion block need confirmatory studies.

Author Contributions RW, SK, and LD wrote the main manuscript text. All authors reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests Jing Wang is founder and scientific board advisory member of Pallas Technologies. The other authors declare no competing interests.

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